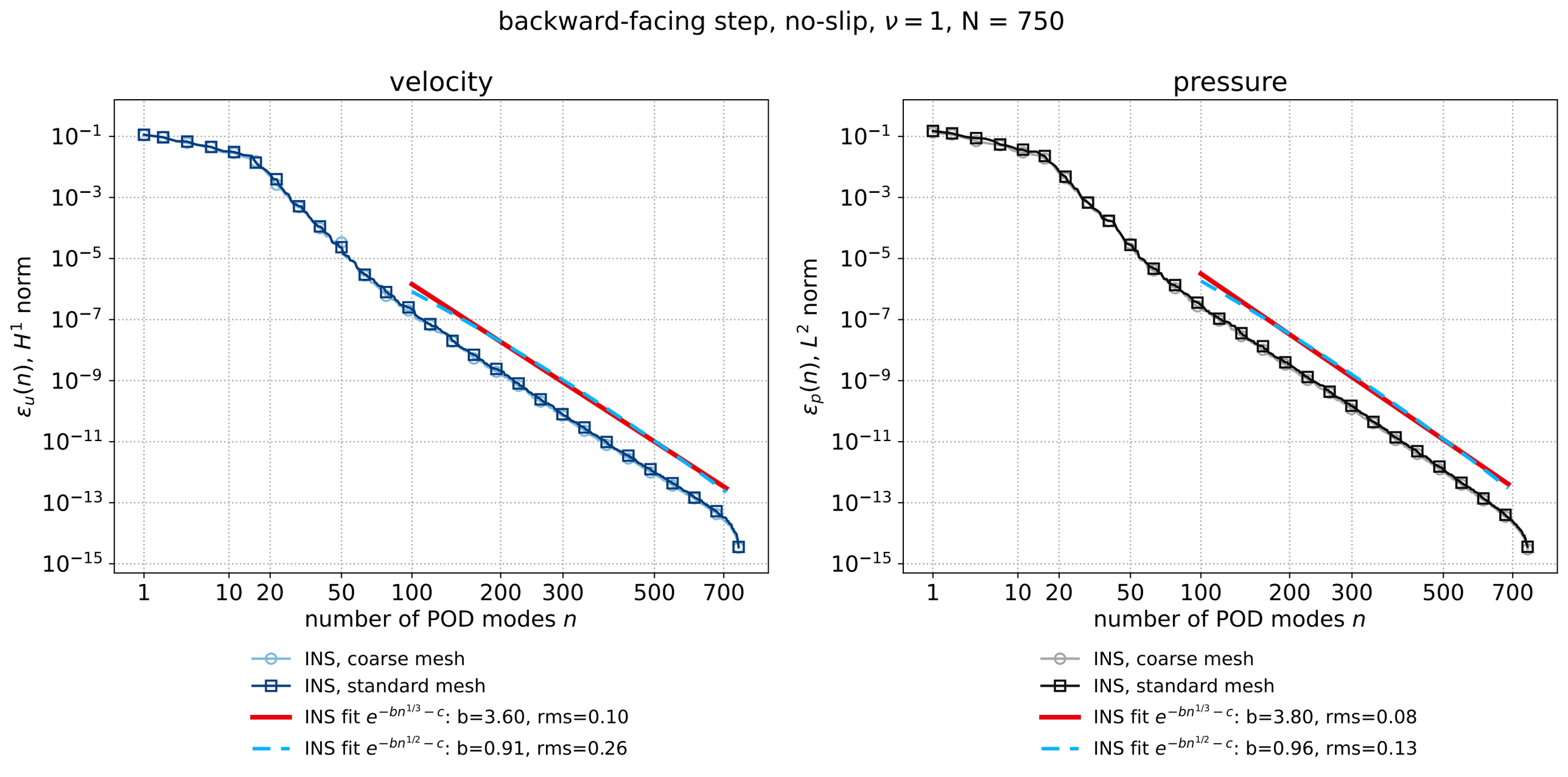
backward-facing step, no-slip, $\nu = 1$, N = 750
velocity
pressure
$\varepsilon_u(n)$, H^1 norm
$\varepsilon_p(n)$, L^2 norm
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{-15}
1
10
20
50
100
200
300
500
700
number of POD modes n
INS, coarse mesh
INS, standard mesh
INS fit $e^{-bn^{1/3}-c}$: b=3.60, rms=0.10
INS fit $e^{-bn^{1/2}-c}$: b=0.91, rms=0.26
INS, coarse mesh
INS, standard mesh
INS fit $e^{-bn^{1/3}-c}$: b=3.80, rms=0.08
INS fit $e^{-bn^{1/2}-c}$: b=0.96, rms=0.13